

Discrete-time Markov Chains

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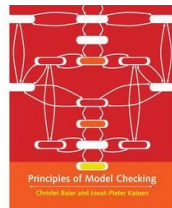
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Selection of slides based on Lecture 2 (Discrete-time Markov Chains) from:

- Dave Parker (DP), “Probabilistic Model Checking”, slides for a lecture course taught in the Department of Computer Science at Oxford University (2011) <http://www.prismmodelchecker.org/lectures/pmc/>

Probabilistic Model Checking

- Formal verification of systems that exhibit probabilistic behaviour
 - e.g. systems with failures/unreliability or randomised algorithms/protocols
- PRISM model checker provides probabilistic operators that take as parameters state and path properties
- In this lecture the concept of a path is introduced formally in the context of Discrete-time Markov chains (DTMCs)
 - The concept of a **path** is employed in all types of model supported by PRISM: DTMCs, CTMCs, MDPs and PTAs
- **Overview** of this lecture
 - Probability basics
 - **Discrete-time Markov chains** (DTMCs: definition, properties, examples)
 - **Formalizing path-based properties** of DTMCs
 - Probability space over infinite paths
- Sources/further reading:



Principles of Model Checking
Christel Baier and Joost-Pieter Katoen
MIT Press, 2008

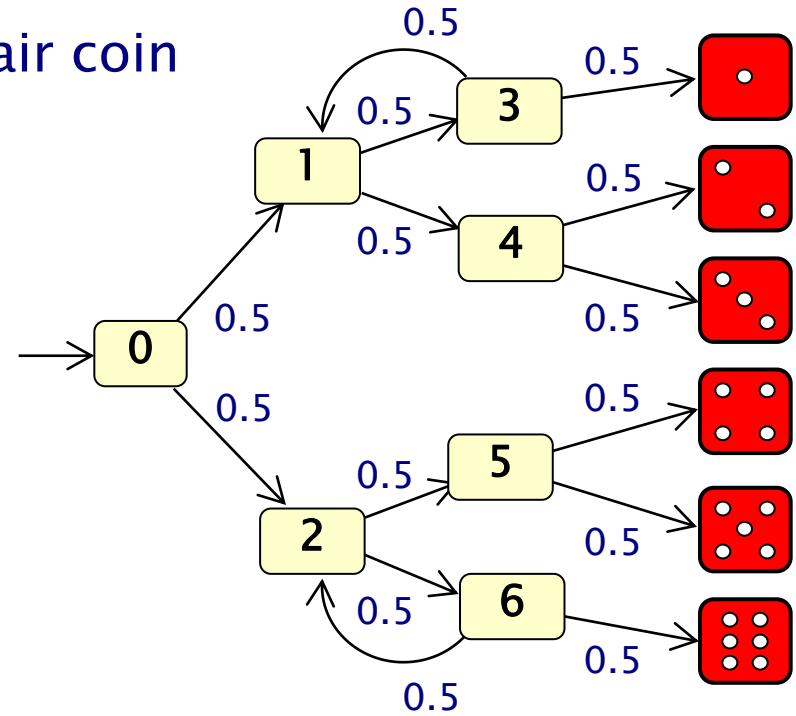
Probability basics

- First, need an experiment
 - The **sample space** Ω is the set of possible outcomes
 - An **event** is a subset of Ω , can form events $A \cap B$, $A \cup B$, $\Omega \setminus A$
- Examples:
 - toss a coin: $\Omega = \{H, T\}$, events: “H” (head), “T” (tail)
 - toss two coins: $\Omega = \{(H, H), (H, T), (T, H), (T, T)\}$,
event: “at least one H”
 - toss a coin ∞ -often: Ω is set of infinite sequences of H/T
event: “H in (either of) the first 3 throws”
- Probability is:
 - $\Pr(\text{“H”}) = \Pr(\text{“T”}) = 1/2$, $\Pr(\text{“at least one H”}) = 3/4$
 - $\Pr(\text{“H in the first 3 throws”}) = 1/2 + 1/4 + 1/8 = 7/8$

Probability example

- **Modelling a 6-sided die using a fair coin**

- algorithm due to Knuth/Yao:
- start at 0, toss a coin
- upper branch when H
- lower branch when T
- repeat until value chosen



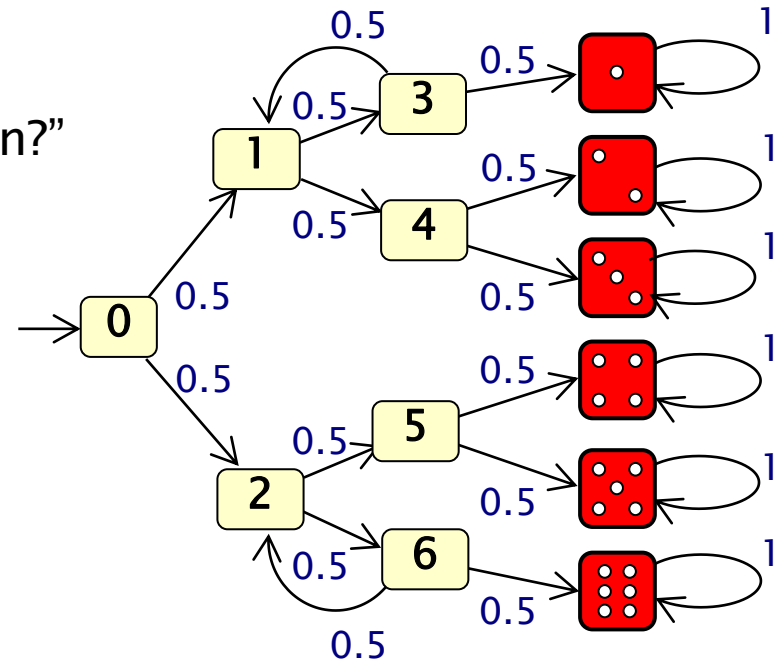
- **Is this algorithm correct?**

- e.g. probability of obtaining a 4?
- Obtain as disjoint union of events
- Execution paths for (event) “eventually 4”: THH, TTTHH, TTTTTHH, ...
- $\Pr(\text{“eventually 4”}) = (1/2)^3 + (1/2)^5 + (1/2)^7 + \dots = 1/6$

- PRISM specification available at <http://www.prismmodelchecker.org/tutorial/die.php>

Example...

- Other properties?
 - “what is the probability of termination?”
- e.g. efficiency?
 - “what is the probability of needing more than 4 coin tosses?”
 - “on average, how many coin tosses are needed?”
- Probabilistic model checking provides a framework for these kinds of properties...
 - modelling languages
 - property specification languages
 - model checking algorithms, techniques and tools



Discrete-time Markov chains

- State-transition systems augmented with probabilities

- States

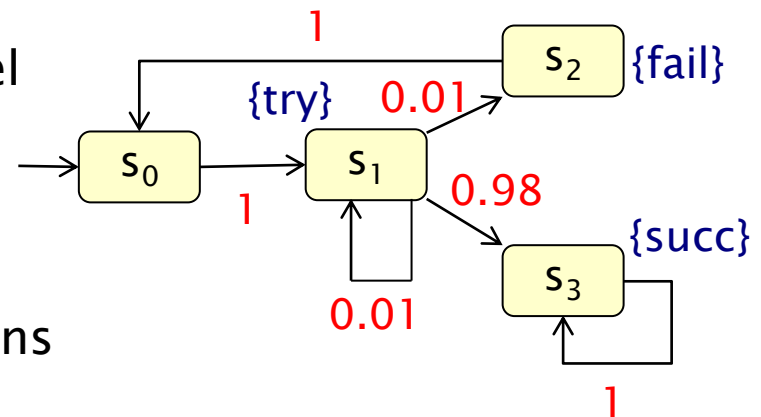
- **set of states** representing possible configurations of the system being modelled

- Transitions

- transitions between states model evolution of system's state; occur in **discrete time-steps**

- Probabilities

- probabilities of making transitions between states are given by **discrete probability distributions**

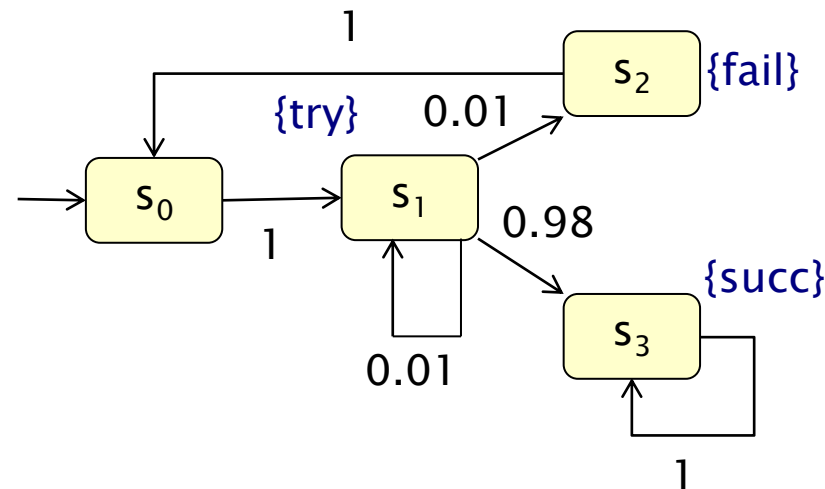


Markov property

- If the current state is known, then the future states of the system are independent of its past states
- i.e. the current state of the model contains all information that can influence the future evolution of the system
- also known as “memorylessness”

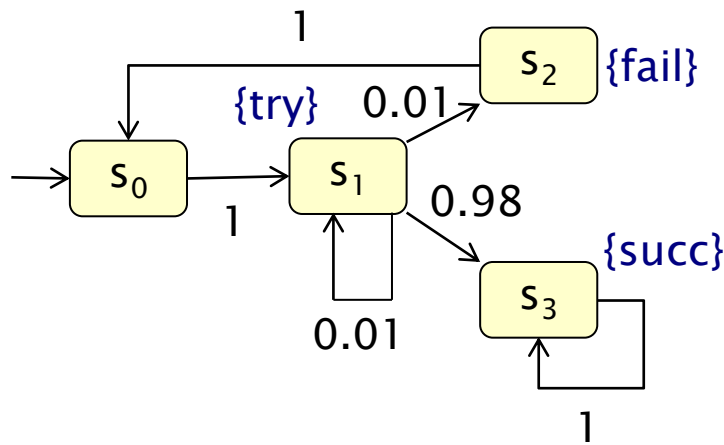
Simple DTMC example

- Modelling a very simple communication protocol
 - after one step, process starts **trying** to send a message
 - with probability 0.01, channel unready so wait a step
 - with probability 0.98, send message **successfully** and stop
 - with probability 0.01, message sending **fails**, restart



Discrete-time Markov chains

- Formally, a DTMC D is a tuple $(S, s_{\text{init}}, P, L)$ where:
 - S is a set of states (“state space”)
 - $s_{\text{init}} \in S$ is the initial state
 - $P : S \times S \rightarrow [0,1]$ is the **transition probability matrix** where $\sum_{s' \in S} P(s, s') = 1$ for all $s \in S$
 - $L : S \rightarrow 2^{\text{AP}}$ is function labelling states with atomic propositions (taken from a set AP)



Simple DTMC example

$$D = (S, s_{\text{init}}, P, L)$$

$$S = \{s_0, s_1, s_2, s_3\}$$

$$s_{\text{init}} = s_0$$

$$AP = \{\text{try}, \text{fail}, \text{succ}\}$$

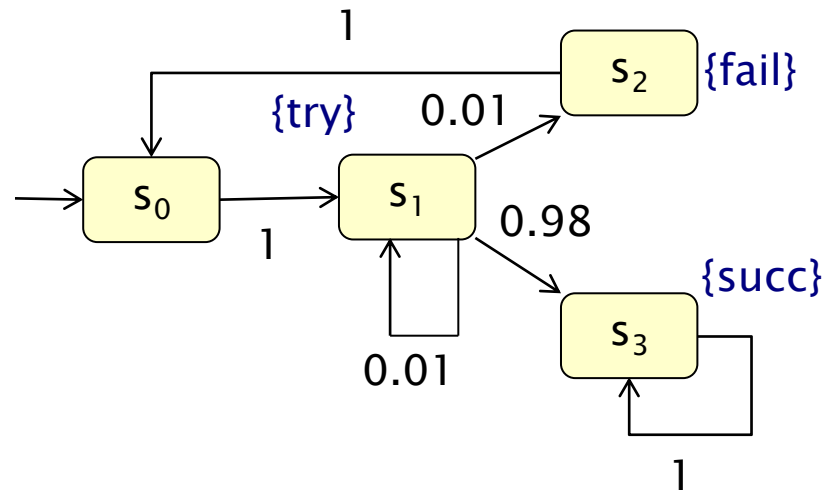
$$L(s_0) = \emptyset,$$

$$L(s_1) = \{\text{try}\},$$

$$L(s_2) = \{\text{fail}\},$$

$$L(s_3) = \{\text{succ}\}$$

$$P = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0.01 & 0.01 & 0.98 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



Some more terminology

- **P** is a **stochastic** matrix, meaning it satisfies:
 - $P(s,s') \in [0,1]$ for all $s,s' \in S$ and $\sum_{s' \in S} P(s,s') = 1$ for all $s \in S$
- An **absorbing state** is a state s for which:
 - $P(s,s) = 1$ and $P(s,s') = 0$ for all $s \neq s'$
 - the transition from s to itself is sometimes called a **self-loop**
- **Note:** Since we assume **P** is stochastic...
 - every state has at least one outgoing transition
 - i.e. no **deadlocks** (in model checking terminology)

Other assumptions made here

- We consider **time-homogenous** DTMCs
 - transition probabilities are independent of time
- We will (mostly) assume that the state space S is **finite**
 - in general, S can be any countable set
- Initial state $s_{\text{init}} \in S$ can be generalised...
 - to an initial probability distribution $s_{\text{init}} : S \rightarrow [0,1]$
- Transition probabilities are reals: $P(s,s') \in [0,1]$
 - but for algorithmic purposes, are assumed to be rationals

Properties of DTMCs

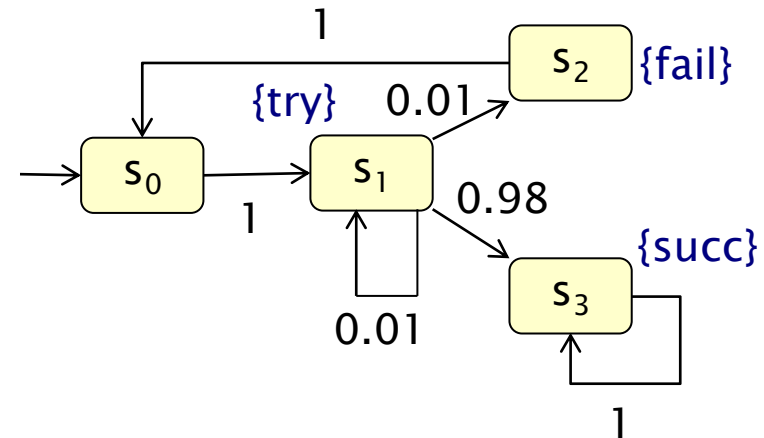
- Path-based properties
 - what is the probability of observing a particular behaviour (or class of behaviours)?
 - e.g. “what is the probability of throwing a 4?”
- Transient properties
 - probability of being in state s after t steps?
- Steady-state
 - long-run probability of being in each state
- Expectations
 - e.g. “what is the average number of coin tosses required?”

DTMCs and paths

- A **path** in a DTMC represents an **execution** (i.e. one possible behaviour) of the system being modelled

- **Formally:**

- infinite sequence of states $s_0s_1s_2s_3\dots$ such that $P(s_i, s_{i+1}) > 0 \ \forall i \geq 0$
- infinite unfolding of DTMC



- **Examples:**

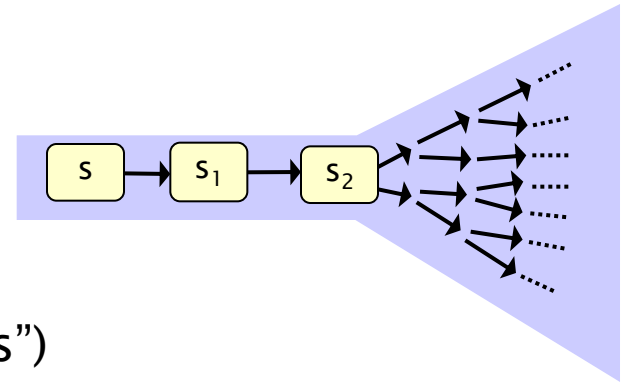
- never succeeds: $(s_0s_1s_2)^\omega$
- tries, waits, fails, retries, succeeds: $s_0s_1s_1s_2s_0s_1(s_3)^\omega$

- **Notation:**

- **Path**(s) = set of all infinite paths starting in state s
- also sometimes use finite (length) paths
- **Path_{fin}**(s) = set of all finite paths starting in state s

Paths and probabilities

- To reason (quantitatively) about this system
 - need to define a **probability space over paths**
- Intuitively:
 - sample space: $\text{Path}(s)$ = set of all infinite paths from a state s
 - events: sets of infinite paths from s
 - basic events: **cylinder sets** (or “cones”)
 - cylinder set $\text{Cyl}(\omega)$, for a finite path ω
= set of **infinite paths with the common finite prefix ω**
 - for example: $\text{Cyl}(ss_1s_2)$



Probability spaces

- Let Ω be an arbitrary non-empty set
- A σ -algebra (or σ -field) on Ω is a family Σ of subsets of Ω ($\Sigma \subseteq 2^\Omega$) closed under complementation and countable union, i.e.:
 - if $A \in \Sigma$, the complement $\Omega \setminus A$ is in Σ
 - if $A_i \in \Sigma$ for $i \in \mathbb{N}$, the union $\bigcup_i A_i$ is in Σ
 - the empty set $\emptyset$ is in Σ
- Elements of Σ are called **measurable sets** or **events**
- Theorem: For any family F of subsets of Ω , there exists a unique smallest σ -algebra on Ω containing F

Probability spaces

- A **probability space** is a triple $(\Omega, \Sigma, \text{Pr})$ where
 - Ω is the sample space
 - Σ is the set of events: $\Sigma (\subseteq 2^\Omega)$ is a σ -algebra on Ω
 - $\text{Pr} : \Sigma \rightarrow [0,1]$ is the **probability measure**:
 - $\text{Pr}(\Omega) = 1$, and
 - $\text{Pr}(\cup_i A_i) = \sum_i \text{Pr}(A_i)$, for all countable sequences of pairwise disjoint events $A_1, A_2, \dots$ ($A_i \in \Sigma, \forall i \in \mathbb{N}, A_i \cap A_j = \emptyset, \forall i \neq j$)

Probability space – Simple example

- Sample space Ω
 - $\Omega = \{1,2,3\}$
- Event set Σ
 - e.g. powerset of Ω
 - $\Sigma = \{ \emptyset, \{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}, \{2,3\}, \{1,2,3\} \}$
 - (closed under complement/countable union, contains $\emptyset$)
- Probability measure $\Pr$
 - e.g. $\Pr(\{1\}) = \Pr(\{2\}) = \Pr(\{3\}) = 1/3$
 - $\Pr(\{1,2\}) = 1/3 + 1/3 = 2/3$, etc.

Probability space – Simple example 2

- Sample space Ω
 - $\Omega = \mathbb{N} = \{ 0, 1, 2, 3, 4, \dots \}$
- Event set Σ
 - e.g. $\Sigma = \{ \emptyset, \text{“odd”}, \text{“even”}, \mathbb{N} \}$
 - (closed under complement/countable union, contains $\emptyset$)
- Probability measure $\Pr$
 - e.g. $\Pr(\text{“odd”}) = 0.5$, $\Pr(\text{“even”}) = 0.5$

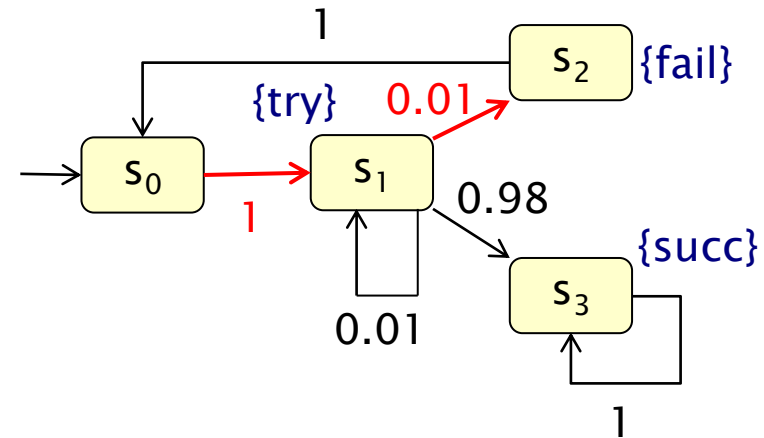
Probability space over paths

- Sample space $\Omega = \text{Path}(s)$ (=set of infinite paths with initial state s)
- Event set $\Sigma_{\text{Path}(s)}$
 - For any finite path ω the **cylinder set** $\text{Cyl}(\omega)$ is
$$\text{Cyl}(\omega) = \{ \omega' \in \text{Path}(s) \mid \omega \text{ is prefix of } \omega' \}$$
 - $\Sigma_{\text{Path}(s)}$ is the **least σ -algebra** on $\text{Path}(s)$ containing $\text{Cyl}(\omega)$ for all finite paths ω starting in s
- Probability measure Pr_s
 - define probability $\text{P}_s(\omega)$ for finite path $\omega = ss_1 \dots s_n$ as:
 - $\text{P}_s(\omega) = 1$ if ω has length one (i.e. $\omega = s$)
 - $\text{P}_s(\omega) = \text{P}(s, s_1) \cdot \dots \cdot \text{P}(s_{n-1}, s_n)$ otherwise
 - define $\text{Pr}_s(\text{Cyl}(\omega)) = \text{P}_s(\omega)$ for any finite path ω
 - Pr_s extends **uniquely** to a probability measure $\text{Pr}_s: \Sigma_{\text{Path}(s)} \rightarrow [0, 1]$
- For further details see
J. Kemeny, J. Snell, A. Knapp. *Denumerable Markov Chains*. Springer, 2nd edition, 1976.

Paths and probabilities – Example

- Paths where sending fails immediately

- $\omega = s_0s_1s_2$
- $\text{Cyl}(\omega) = \text{all paths starting } s_0s_1s_2\dots$
- $\mathbf{P_{s_0}(\omega) = P(s_0,s_1) \cdot P(s_1,s_2)}$
 $\quad = \mathbf{1 \cdot 0.01 = 0.01}$
- $\mathbf{Pr_{s_0}(\text{Cyl}(\omega)) = P_{s_0}(\omega) = 0.01}$



- Paths which are eventually successful and with no failures

- $\text{Cyl}(s_0s_1s_3) \cup \text{Cyl}(s_0s_1s_1s_3) \cup \text{Cyl}(s_0s_1s_1s_1s_3) \cup \dots$
- $\mathbf{Pr_{s_0}(\text{Cyl}(s_0s_1s_3) \cup \text{Cyl}(s_0s_1s_1s_3) \cup \text{Cyl}(s_0s_1s_1s_1s_3) \cup \dots)}$
 $\quad = \mathbf{P_{s_0}(s_0s_1s_3) + P_{s_0}(s_0s_1s_1s_3) + P_{s_0}(s_0s_1s_1s_1s_3) + \dots}$
 $\quad = \mathbf{1 \cdot 0.98 + 1 \cdot 0.01 \cdot 0.98 + 1 \cdot 0.01 \cdot 0.01 \cdot 0.98 + \dots}$
 $\quad = \mathbf{0.9898989898\dots}$
 $\quad = \mathbf{98/99}$

Final Remarks

- Discrete-time Markov chains (DTMCs)
 - State-transition systems augmented with probabilities
- Path-based properties of DTMCs
 - Formalized by defining a probability space over infinite paths
- The concept of a **path** is employed in all types of model supported by PRISM: DTMCs, CTMCs, MDPs and PTAs
- **PRISM** is a tool for quantitative modeling and analysis
 - It provides **probabilistic operators**
 - The parameters are state, **path** and **reward properties**
 - The tool can compute (mean or) **expected values of random variables**
 - Defined based on **reward structures**
 - Example: expected reward/cost cumulated in k steps – the number of steps is counted along (execution) **paths**